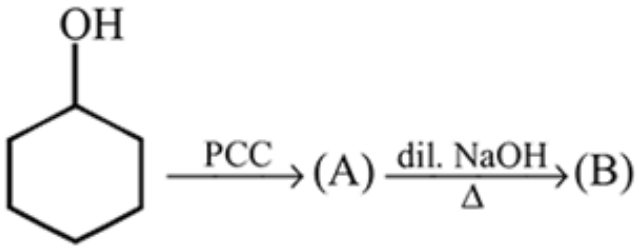
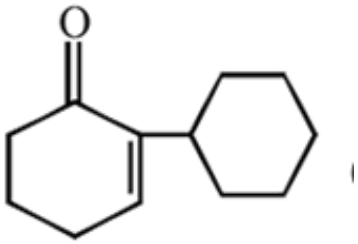
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

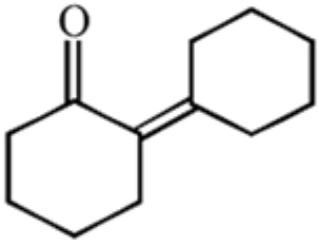


Product (B) is:

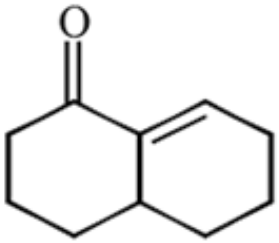
(A)



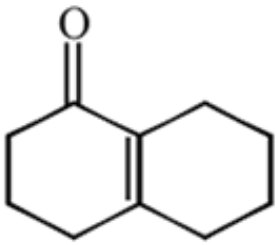
(B)



(C)

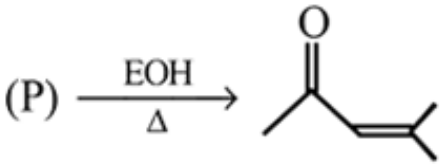


(D)



Q.2

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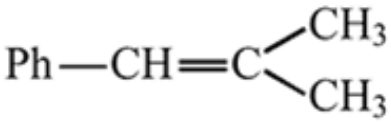


(Q) $\xrightarrow[\Delta]{EOH} Ph - CH_2 - OH + Ph - CO_2$

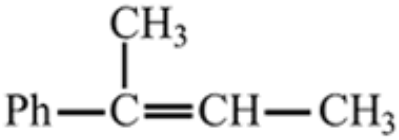
(R) $\xrightarrow{O_3}$ P + Q, Structure of (R) is:

(A) $Ph - CH = CH - CH_3$

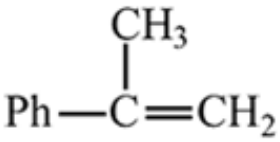
(B)



(C)

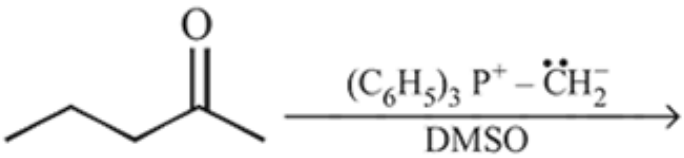


(D)



Q.3

What is the product of the following reaction?



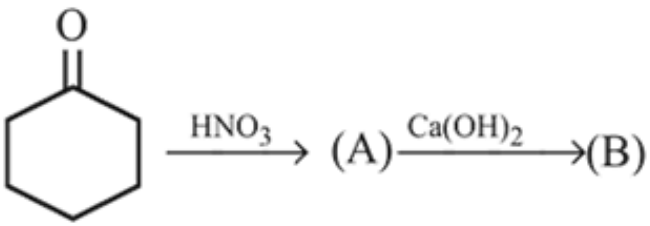
(A) 2 - methyl - 1- pentene

(B) 4 - methyl - 1 - pentene

(C) 2 - methyl - 2 - propyloxirane

(D) 1 - pentene

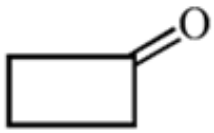
Q.4



Product (B) in this reaction is:

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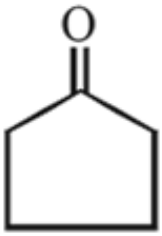
(A)



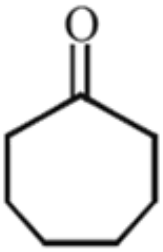
(B)



(C)



(D)



Q.5

In the reaction sequence

$2CH_3CHO \xrightarrow{OH^-} A \xrightarrow{\Delta} B$; the product B is:

(A) $CH_3 - CH_2 - CH_2 - CH_2 - OH$

(B) $CH_3 - CH = CH - CHO$

(C) $CH_3 - CH_2 - CH_2 - CH_3$

(D)

$$CH_3 - \overset{\overset{O}{\|}}{C} - CH_3$$

Q.6

Which one of the following reactions will not result in the formation of carbon-carbon bond?

(A) Reimer-Tieman reaction

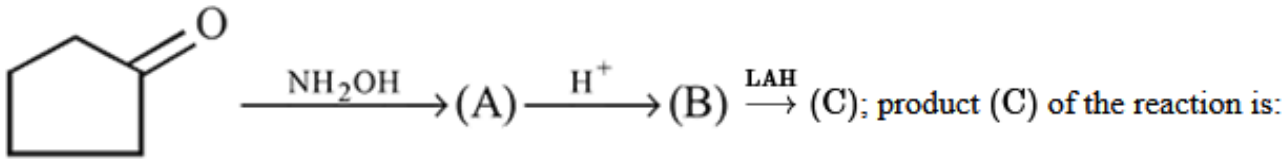
(B) Friedel Craft's acylation

(C) Wurtz reaction

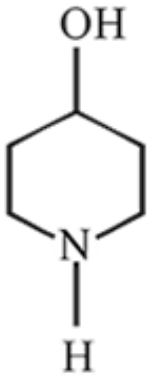
(D) Cannizzaro reaction

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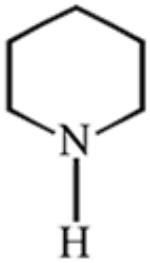
Q.7



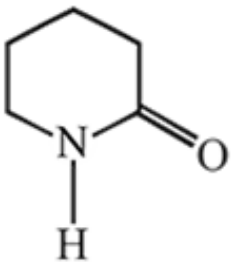
(A)



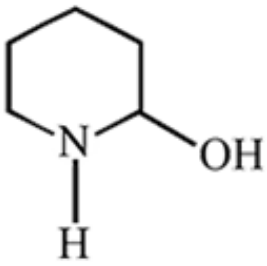
(B)



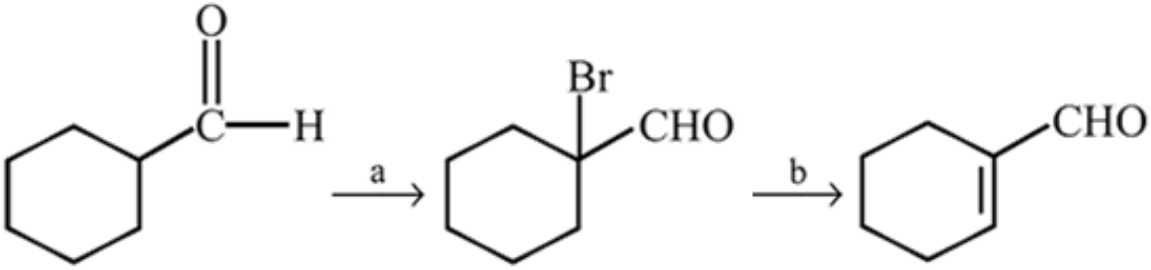
(C)



(D)



Q.8



Identify appropriate reagents for the above reaction:

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(A) $a = Br_2/CCl_4, \quad b = aq.KOH$

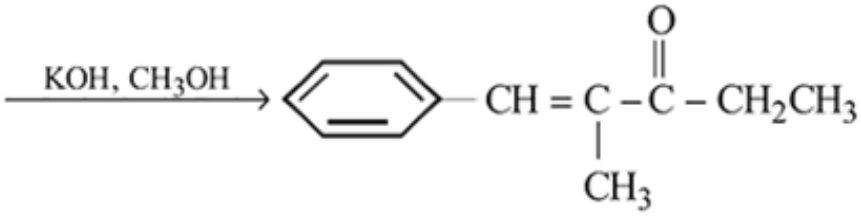
(B) $a = Br_2/H^+, \quad b = aq.KOH$

(C) $a = Br_2/H^+, \quad b = alc.KOH$

(D) $a = Br_2/HO^-, \quad b = aq.KOH$

Q.9

The starting compounds needed to make the following compound?



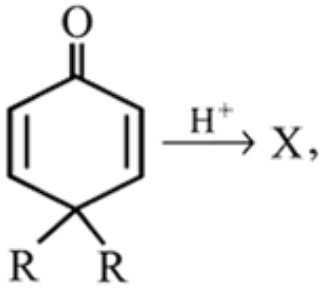
(A) benzaldehyde and 3 -pentanone

(B) acetophenone and 2-butanone

(C) acetophenone and butanal

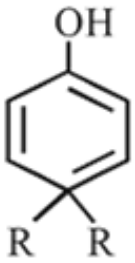
(D) benzaldehyde and 2-pentanone

Q.10

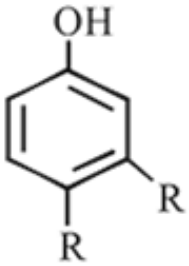


most likely the compound X is

(A)

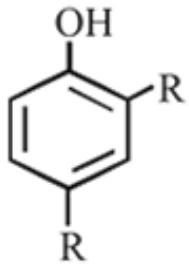


(B)

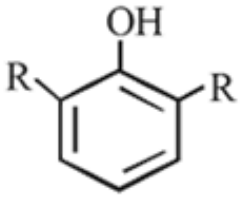


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(C)

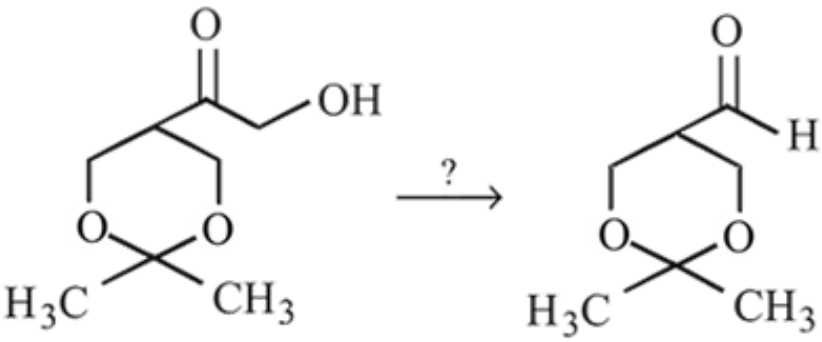


(D)



Q.11

The reagents employed to carry the following transformation



(A) $LiAlH_4, H_2SO_4/$ heat

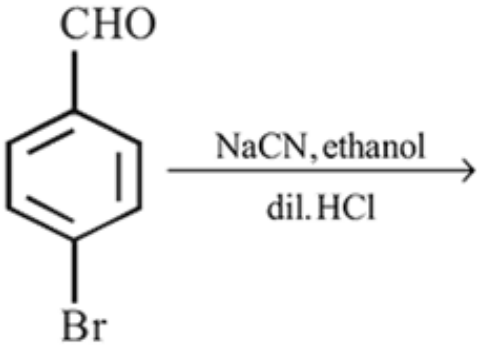
(B) PCC/CH_2Cl_2 followed by HIO_4

(C) $NaBH_4/CH_3OH$ followed by HIO_4

(D) O_3 followed by $(CH_3)_2$ S

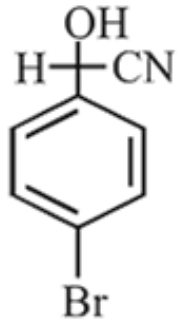
Q.12

What is the product of the following reaction?

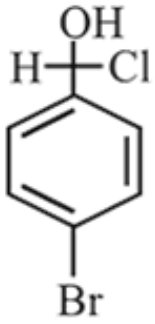


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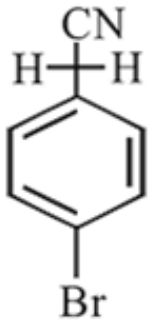
(A)



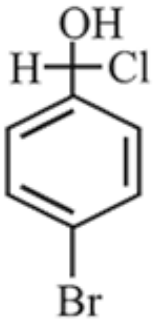
(B)



(C)

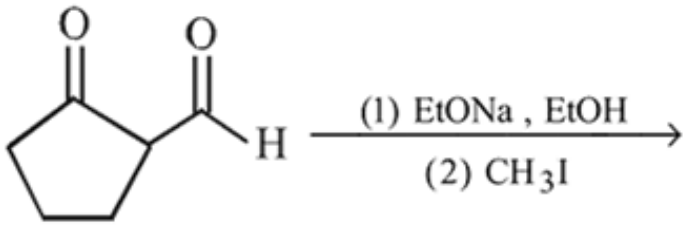


(D)

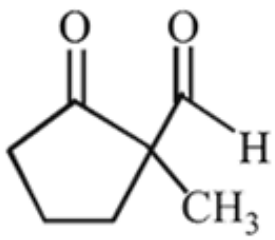


Q.13

What is the product of the following reaction?

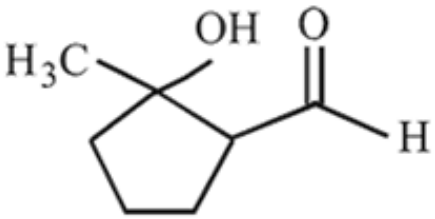


(A)

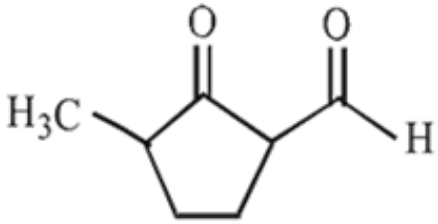


@IITJEE_Advanced

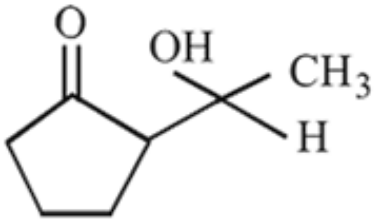
(B)



(C)



(D)



Q.14

Tischenko reaction is a modification of:

(A) Aldol condensation

(B) Claisen condensation

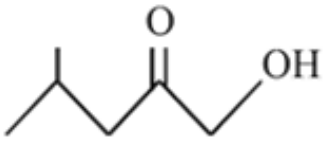
(C) Cannizzaro reaction

(D) Pinacol-pinacolon reaction

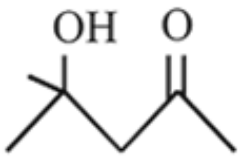
Q.15

Which of the following is the product of aldol condensation?

(A)



(B)

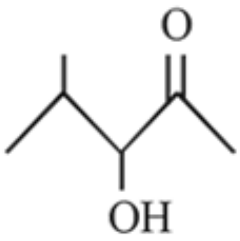


(C)



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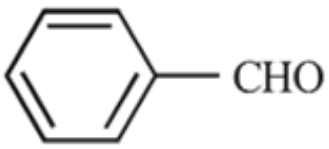
(D)



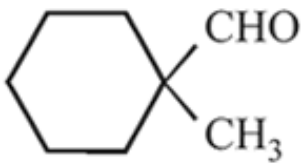
Q.16

Cannizaro's reaction is not given by:

(A)



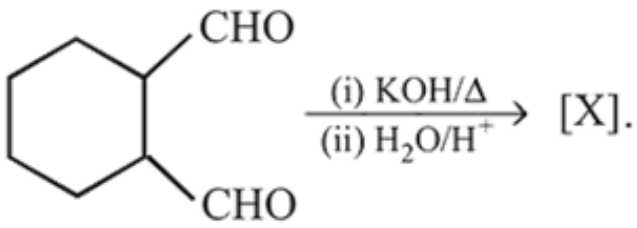
(B)



(C) CH_3CHO

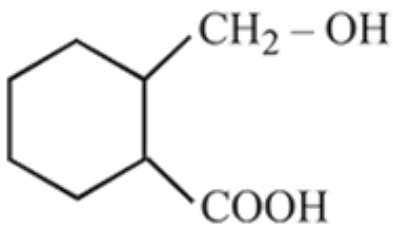
(D) HCHO

Q.17

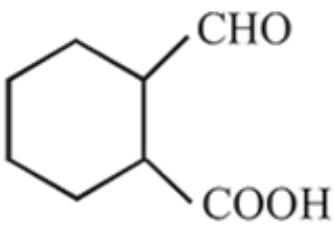


What is $[X]$?

(A)

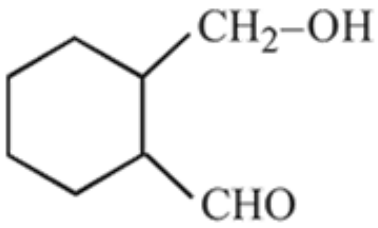


(B)

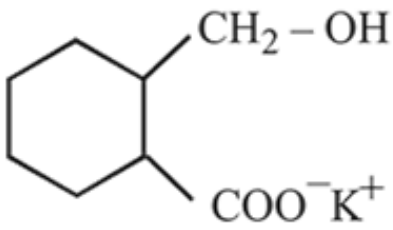


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(C)

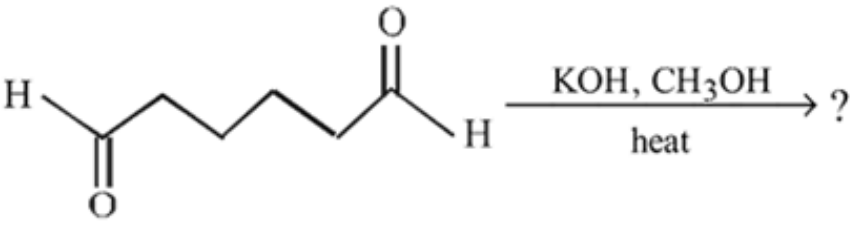


(D)

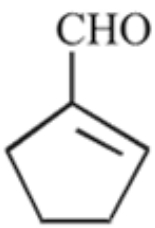


Q.18

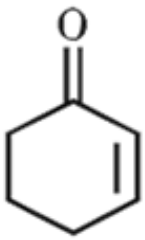
What is the product of the following reaction.



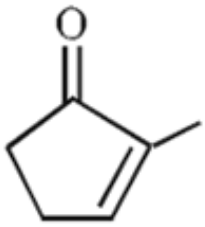
(A)



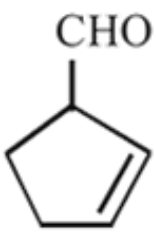
(B)



(C)



(D)



Q.19

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The most reactive compound towards formation of cyanohydrin on treatment with KCN followed by acidification is

(A) benzaldehyde

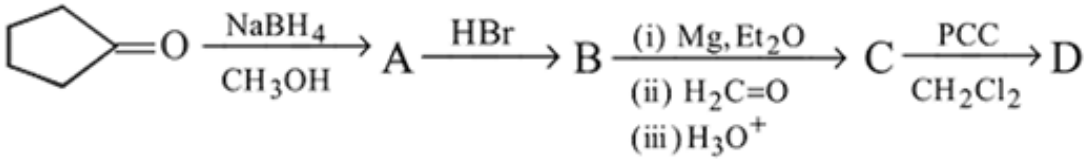
(B) p -nitrobenzaldehyde

(C) phenylacetaldehyde

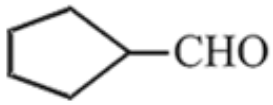
(D) p -hydroxybenzaldehyde

Q.20

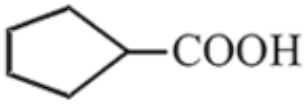
What is D in the following sequence of reactions?



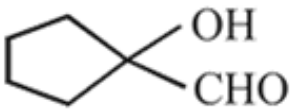
(A)



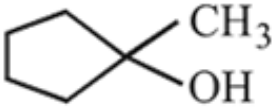
(B)



(C)

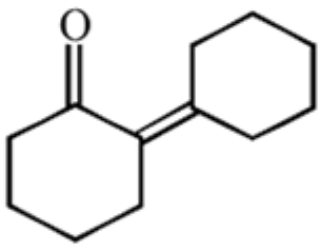


(D)



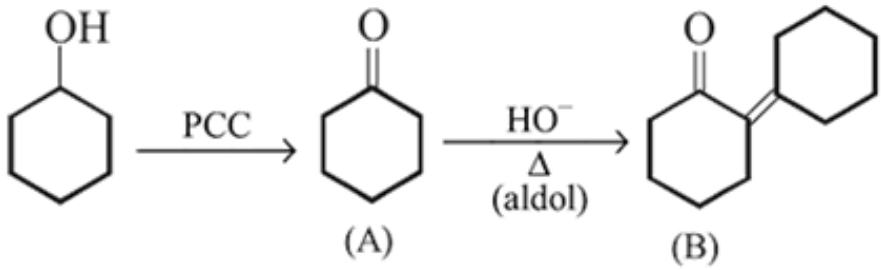
Answers & Solutions

Q.1 Answer:

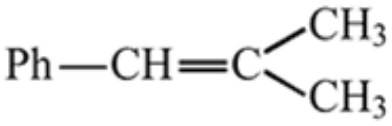


Solution:

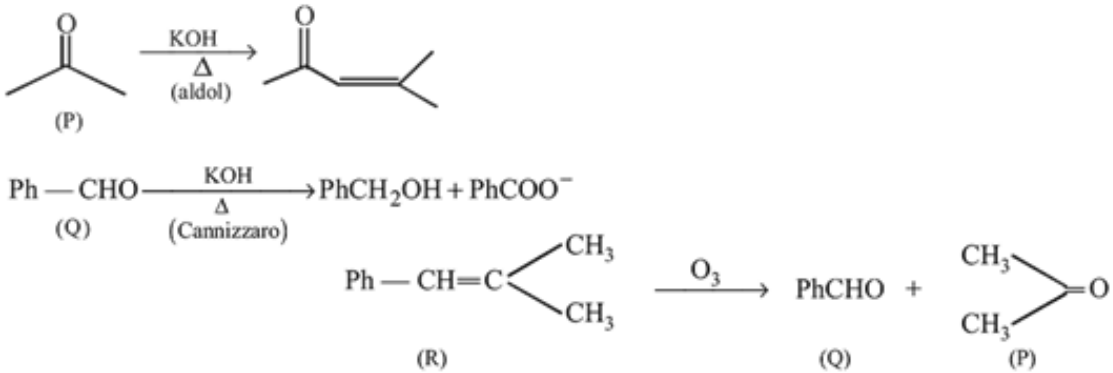
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Q.2 Answer:



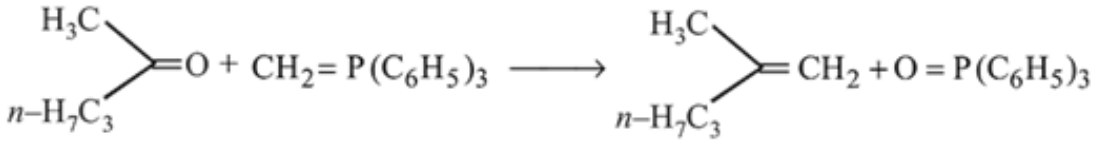
Solution:



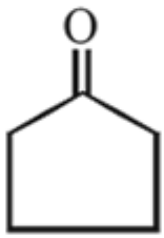
Q.3 Answer:

2 - methyl - 1- pentene

Solution:

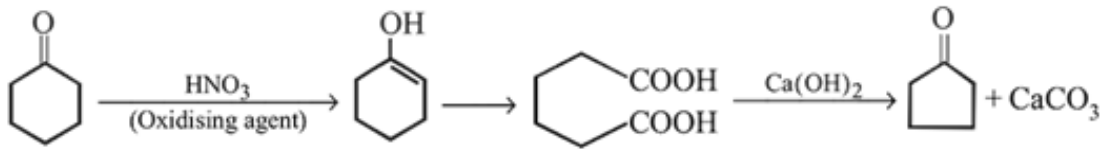


Q.4 Answer:



Solution:

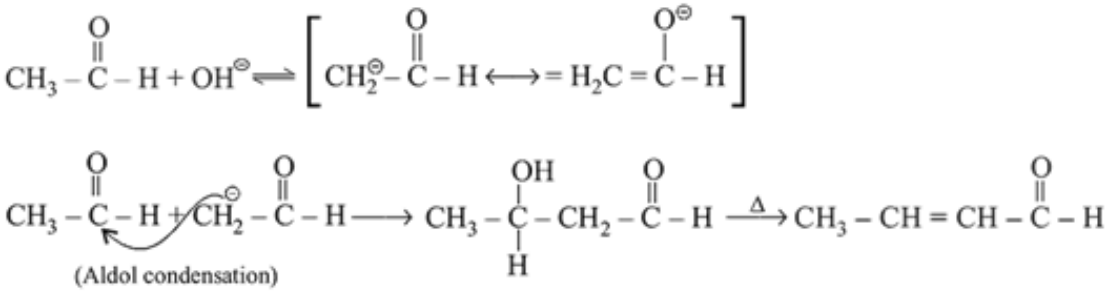
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Q.5 Answer:

$CH_3 - CH = CH - CHO$

Solution:



Q.6 Answer:

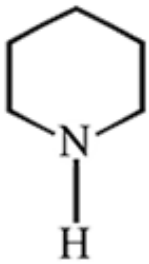
Cannizzaro reaction

Solution:

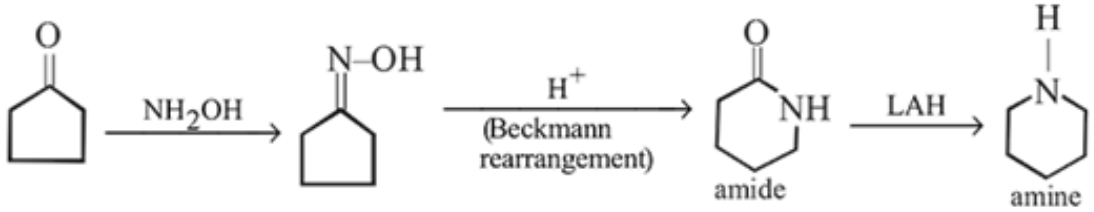
Canizzaro's reaction is a disproportionation reaction of aldehyde in which one molecule of aldehyde is reduced to alcohol whereas other is oxidised to salt of carboxylic acid

$C_6H_5CHO + KOH \longrightarrow C_6H_5CH_2OH + C_6H_5COOK$

Q.7 Answer:



Solution:

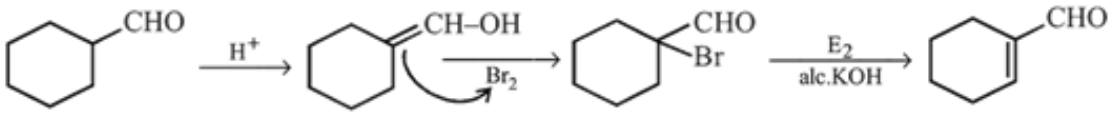


Q.8 Answer:

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$a = Br_2/H^+, \quad b = alc.KOH$

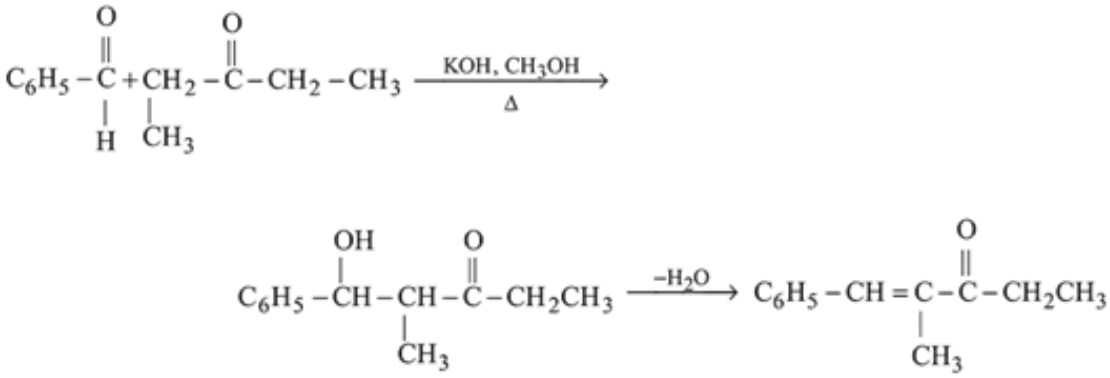
Solution:



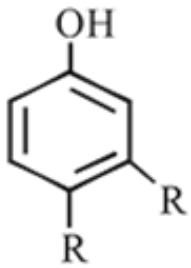
Q.9 Answer:

benzaldehyde and 3 -pentanone

Solution:



Q.10 Answer:



Solution:

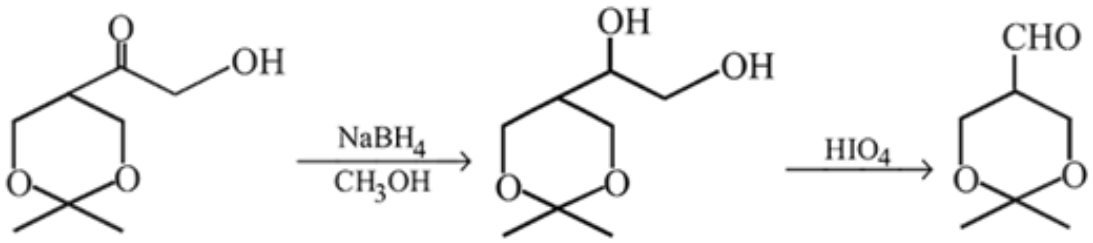
Reaction of 4,4 -dialkylcyclohexadienone with an acid to form phenol with the migration of one of the alkyl groups to the adjacent carbon is known as dienone-phenol rearrangement.

Q.11 Answer:

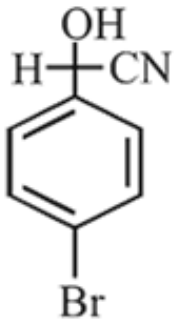
$NaBH_4/CH_3OH$ followed by HIO_4

Solution:

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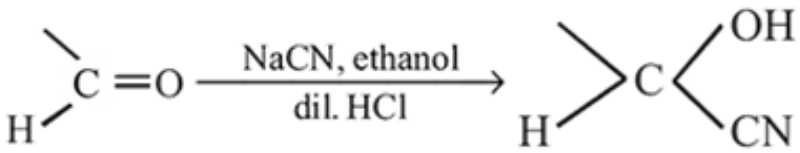


Q.12 Answer:

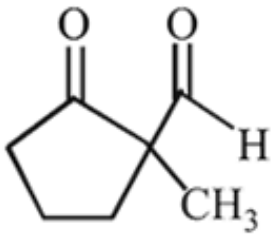


Solution:

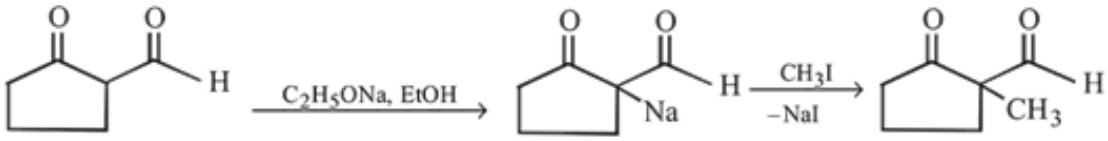
Only-CHO group reacts with CN^- ion and the reaction is



Q.13 Answer:



Solution:



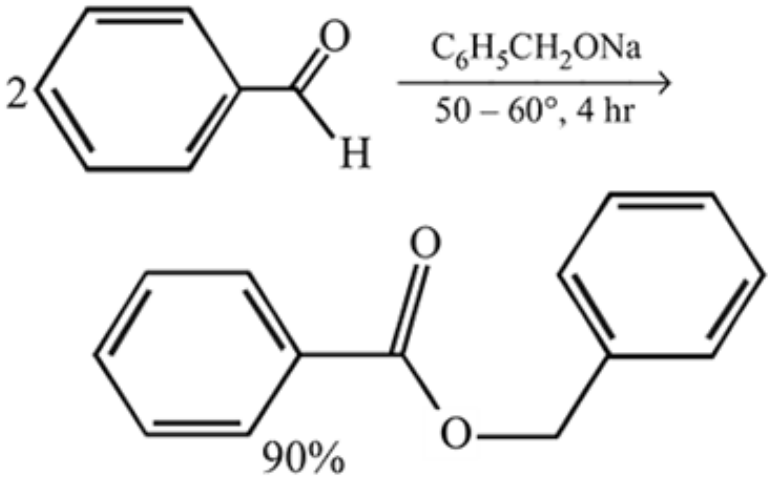
Q.14 Answer:

Cannizzaro reaction

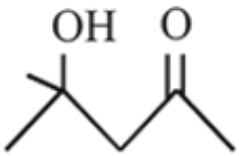
Solution:

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Tishchenko reaction is a modification of Canizzaro's reaction. This reaction involves disproportionation of an aldehyde lacking a hydrogen atom in the alpha position in the presence of an alkoxide. The reaction product is an ester. Catalysts are aluminium alkoxide or sodium alkoxide. In Cannizzaro's reaction the base is sodium hydroxide and the oxidation product is a carboxylic acid and the reduction product is an alcohol.



Q.15 Answer:



Solution:

When two aldehydes or ketones (or carbonyl compounds) having $\alpha-$hydrogen atom reacts in the presence of a base, they result in the formation of $\beta-$hydroxy aldehyde or ketone, called the aldol condensation.

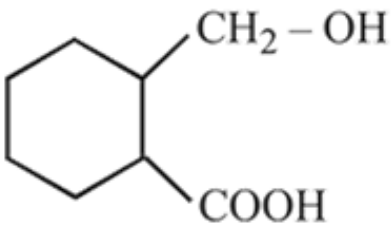
Q.16 Answer:

CH_3CHO

Solution:

Only those aldehydes which do not have α -H atom undergo Cannizaro's reaction. Hence CH_3CHO will not undergo Cannizaro's reaction as it has $3\alpha H$ atoms.

Q.17 Answer:

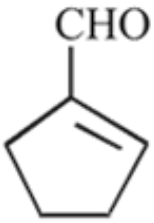


Solution:

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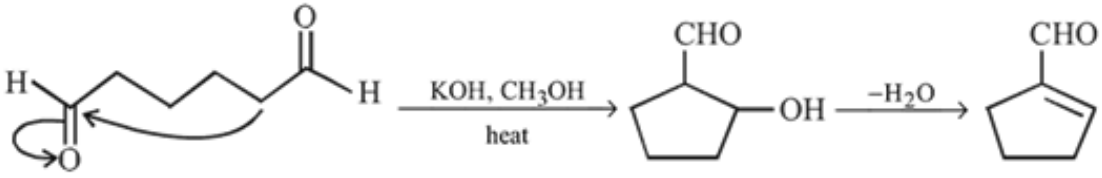
It is an example of intramolecular cannizzaro reaction.

Q.18 Answer:



Solution:

It is an example of intramolecular aldol condensation



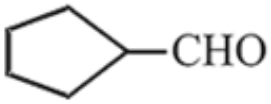
Q.19 Answer:

p -nitrobenzaldehyde

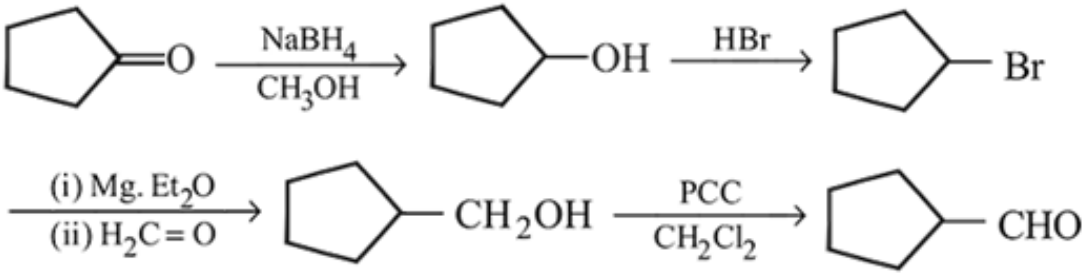
Solution:

$-NO_2$ group is electron withdrawing which enchances the addition of CN^- ion (a nucleophili additon reaction).

Q.20 Answer:



Solution:



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